MILITARY
SPECIFICATION

MIL-R-13963A (SigC)
23 September 1958
Superseding
MIL-R-13963 (SigC)
8 February 1955

RADIO SET AN/TRC-29 ( )
(COMPONENTS OF)

1. SCOPE

1.1 This specification covers the following components of one type of equipment designated as Radio Set AN/TRC-29, used in rear area radio relay systems, operating in the 1700 to 2400 **MISSING WORD** megacycle microwave region, capable of transmitting and receiving multi-channel voice circuits or wide band signals from suitable modulators:

Transmitter Rack consisting of:

Radio Transmitter T-303 ( )/G
Order Wire Transmitter T-389 ( )/TRC-29
Power Supply PP-764 ( )/G

Receiver Rack consisting of:

Radio Receiver R-418 ( )/G
Order Wire Receiver R-543 ( )/TRC-29
Power Supply PP-690 ( )/G
Power Supply PP-689 ( )/G

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, of the issue in effect on date of invitations for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

LLL-F-291	Fiberboard; corrugated, single face (flexible)
PPP-B-621	Boxes, wood, nailed and lock-corner
PPP-B-636	Boxes, fiber

MILITARY

MIL-B-13239	Barrier material, waterproofed, flexible, all temperature
MIL-P-75	Packaging, packing and container marking of electron tubes
MIL-P-116	Preservation, methods of
MIL-B-117	Bags, Interior, Packaging
MIL-T-152	Treatment moisture and fungus-resistant, of communications, electronic, and associated electrical equipment.
MIL-L-10547	Liners, case, waterproof
MIL-P-11268	Parts, materials, and processes used in Signal Corps Equipment
MIL-I-11748	Interference, reduction for electrical and electronic equipment.
MIL-M-13231	Marking of electronic items
MIL-F-14072	Finishes for ground Signal Equipment

FSC 5820

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STANDARDS

MILITARY

MIL-STD-105	Sampling procedures and tables for inspection by attributes
MIL-STD-129	Marking for shipment and storage
MIL-STD-169	Extreme-temperature cycle
MIL-STD-170	Moisture resistance test cycle for ground Signal Equipment
MIL-STD-252	Wired equipment, classification of visual and mechanical defects

DRAWINGS

SIGNAL CORPS

SC-0-33073	Form to be used for Pilot Pack Process

2.2 Other publications - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification No. A-3 and No. 14

American Trucking Association, Inc.
1424 Sixteenth Street
Washington 6, D C

ASSOCIATION OF AMERICAN RAILROADS

Uniform Freight Classification No. 4

Uniform Freight Classification
202 Union Station
Chicago 6, Illinois

Consolidated Freight Classification Rules No. 21

Consolidated Classification Committee
1 Park Avenue and 33rd Street
New York 16, New York

3. REQUIREMENTS

3.1 Procurement model.- The procurement model shall be the one which will be available for inspection by prospective bidders and will be lent to the contractor, in accordance with the invitation for bids. Unless otherwise specified herein or in the invitation for bids, the equipment shall conform to the following:

3.1.1 Construction.- Physical construction of the equipment shall conform to the procurement model.

3.1.2 Features.- The equipment shall incorporate all features of the procurement model.

3.1.3 Verification.- As soon as the procurement model is received, and with the minimum necessary adjustment and alinement (if any), the contractor shall measure its performance for compliance with specified characteristics of the equipment on contract. If the model does not meet the specified performance characteristics, the contractor shall adjust or aline it for optimum performance and then repeat the measurements. The contractor shall immediately prepare copies of a test report on the measurements, and forward them as required by the contract. In case of conflict between specified performance characteristics and performance of the model, the former shall govern.

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3.2 Parts, materials, and processes general.- In addition to any requirements of this specification, covering parts, materials, and processes, such items shall conform to Specification MIL-P-11268 including the selection requirements therein.

3.3 Marking.-

3.3.1 General.- Marking shall conform to Specification MIL-M-13231. Front panel markings shall conform to group I.

3.3.2 Visibility.- Wherever practicable parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.3.3 Serial numbers.- All components listed in 1.1, shall be serial numbered.

3.4 Tropicalization.-

3.4.1 Treatment of assembled equipment.- Assembled equipment shall be treated in accordance with Specification MIL-T-152 unless both of the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc) which is applicable on contract.
(2) It does not incorporate any material listed in 3.4.2 unless the material has been given the treatment specified in Specification MIL-P-11268 or a treatment approved by the contracting officer.
(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL-F-14072.

3.4.2 Materials not resistant to moisture and fungi.- The following materials are considered not resistant to moisture and fungi:

(a) Cellulose, regenerated.
(b) Cotton
(c) Cork
(d) Felt, hair or wool
(e) Jute
(f) Leather
(g) Linen
(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.
(i) Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base.

3.4.3 Statement of treatment.- The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. This statement shall include the statements required by Specifications MIL-P-11268 and MIL-T-152, as applicable.

3.5 Power Input.- Radio Set AN/TRC-29 requires a single-phase AC input. The set draws 2.3 kilovolt-amperes at a power factor of better than 90 percent at a line voltage of 115. The voltage supply available to the radio set should be between 100 volts and 130 volts or between 200 and 260 volts at a frequency of 50 to 60 cps.

3.6 Radio interference suppression.- The equipment shall conform to the requirements of Specification MIL-I-11748 and the interference limits set forth therein, with the antenna of the interference measuring equipment located 5 feet from the equipment under test.

3.7 Service Conditions.- The equipment shall meet the following service condition:

3.7.1 Equipment nonoperating - Except as stated in 3.7.1 (b), the equipment shall comply with

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the operational requirements of 3 7.2 after subjection to any of the following nonoperating conditions successively or in combinations encountered during world wide short-term storage and transit:

(a) Temperature.- Exposure in the range of + 160°F to -80°F; however, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature not to exceed 24 hours at any one time.

(b) Relative Humidity.- Relative humidity up to 97 percent, for an indefinite period of time; and exposure at 100 percent relative humidity, with condensation, for 4 hours. Following subjection to such conditions and a power-on period of 16 hours at normal room temperature and relative humidity, the equipment shall meet the requirements of 3.7.2.

(c) Elevation.- Elevation up to 25,000 feet above sea level.

(d) Vibration, bounce, and shock.- As described in 3.11 and 3.12

3.7.2 Equipment operating.- After a warmup period of one hour, the equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations:

(a) Operation.- Continuous operation for a period of 1 year, alternating 8 hours on and 16 hours off, with no more than normal maintenance and replacement of parts. Compliance will be determined by examination of the preproduction sample.

(b) Temperature.- Ambient temperature in the range of + 132°F to -40°F. Exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours.

(c) Relative humidity.- Relative humidity up to 97 percent; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

(d) Elevation.- Elevation up to 10,000 feet above sea level.

3.8 Transmitter Rack.- The transmitter rack consists of one Transmitter, Radio T-303( )/G; one Transmitter, Order Wire, T-389( )/TRC-29; one Power Supply PP-764( )/G, mounted in Case, Standardized Components, Electrical CY-1280 ( )/U and internal interconnecting cables.

3.8.1 Transmitter T-303( )/G.- The radio transmitter shall have the following characteristics.

3.8.1.1 Tuning.- Transmitter, Radio T-303( )/G shall be continuously tunable over the frequency range of 1700 to 2400 mcs. and with automatic Frequency control crystals shall operate on the following 27 pre-set channels in the 1700 to 2400 mc range:

Channel Number	Channel Frequency -mc	Transmitter Crystal Cr-52/U or Cr-54/U	Channel Number	Channel Frequency -mc	Transmitter Crystal CR-52/U or CR-54/
1	1715	76.458	15	2065	72.033
2	1745	77.708	16	2090	73.667
3	1775	78.958	17	2115	74.500
4	1805	80.208	18	2140	75.333
5	1835	81.458	19	2165	76.167
6	1860	82.500	20	2190	77.000
7	1885	83.542	21	2215	77.833
8	1910	67.667	22	2245	78.833
9	1935	68.500	23	2275	79.833
10	1960	69.333	24	2310	81.000
11	1980	70.000	25	2335	81.833
12	2000	70.667	26	2360	82.667
13	2020	71.333	27	2385	83.500
14	2040	72.000			

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3.8.1.2 Modulator Gain.- When tested as specified in 4.7.2 the gain of the modulator shall be at least 33.0 V. RMS.

3.8.1.3 Modulator Response.- When tested as specified in 4.7.3 the modulator response shall be as follows:

at 250 KC	+ .5 db) -3.0 db)	from 100 KC Reference
at 7 MC	+ .5 db) -3.0 db)	from 100 KC Reference

3.8.1.4 Monitor Receiver Output Level.- The output level shall be 2000 mv minimum when measured in accordance with 4.7.4.

3.8.1.5 Sawtooth Linearity.- The sawtooth pattern shall show no irregularities.

3.8.1.6 Output Power.- With an input of 115 volts, the power output shall be as follows:

1715 MCS - 8 watts Minimum
2000 MCS - 6 " "
2190 MCS - 5 " "
2385 MCS - 4 " "

with line voltage decreased to 103.5 volts the power output shall be at least .63 times the readings obtained at 115 volts.

3.8.1.7 AFC Hunting.- The AFC system shall return the AFC meter readings to approximately 0 with no hunting with a maximum of 4 reversals of the tuning motor.

3.8.1.8 Monitor Receiver Video Response.- When tested as specified in 4.7.8, the Receiver Video Response shall be as follows.

at 300 KC	-3 db) + 1 db)	from 100 KC Reference
at 4.5 MC	-3 db) + 1 db)	from 100 KC Reference

3.8.1.9 Order Wire Alarm.- With a reference obtained at 1.5 mc and a 6 mc signal set for 15% of the reference, the transmitter operating lamp shall be lit. With the generator level reduced as outlined in 4.7.9, the voltage shall be a minimum of 2% of the reference reading.

3.8.1.10 Metering.- Following requirements shall be met when the meter switches are set in the following positions:

Left Hand Meter	Allowable
AFC 1ST MULT GRID	30-140
AFC MIXER	10-100
KLYSTRON POWER OUT	20-145
PA CATH CUR	100 Absolute Max. = 125
1ST LIM GRID	10 (Minimum)
AFC CORR +	0 + 5
AFC CORR -	0 ± 5
REFL V	70-100
BEAM V	100 ± 5

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Right Hand Meter

AFC 2ND MULT GRID	20-130	
WAVEMETER	30-145	
PA GRID CUR	25-140	
PA PLATE V	50-130	Approx. when cathode current reading = 100
+ 300V	90-105	
+ 250V	90-104	
+ 150V	93-109	
-150V	95-105	

3.8.2 Transmitter Order Wire T-389( )/TRC-29.- The Order Wire Transmitter shall have the following characteristics:

3.8.2.1 Frequency Drift VS. Line.- The carrier frequency when checked with a counter shall be as follows:

VARIAC ADJUSTMENT	FREQUENCY MEASUREMENT	
115 V AC	6.000 MCS	± 1KC
103.5 V AC	6.000 MCS	± 2 KC
126.5 V AC	6.000 MCS	± 2 KC

3.8.2.2 Ripple Voltage vs Line Voltage - The ripple voltage shall be less than 30 millivolts when measured as outlined in 4.8.2

3.8.2.3 Deviation O.W. Transmitter R.F.- The transmitter frequency deviation shall be at least ± 50 KC for a 0.3 volt, 20.8 KC signal.

3.8.2.4 Output Voltage.- The output voltage as measured with a VTVM shall be a minimum of 350 millivolts RMS.

3.8.2.5 Frequency Response of Audio Amplifier.- The THRU input shall have an output of at least .225 V and not greater than .300 V with an input signal of 300 cps, and an output of at least 0.280 V and not greater than .335 V with a 3000 cps input signal. The LOCAL input shall have an output of at least .225 V and not greater than .300 V when a signal of 300 cps is applied. With an input signal of 3000 cps the output shall be at least .280 V and not greater than .335V.

3.8.2.6 Alarm Circuits.- When the 6 MC ALARM button is depressed, the alarm light shall light, and by depressing the OFF FREQUENCY ALARM button the light shall light.

3.8.2.7 Metering Circuits. Following requirements shall be met when the meter switches are

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set in the following position:

S-2 Position	S-1 Position	Allowable
CARR LIM GRID	OPR	75 - 135
DISC +	OPR	0 ± 10
AUDIO	OPR	Adjust R24 for 100
DISC-	OPR	0 ± 10
AUDIO	THRU TONE	50-140
AUDIO	LOCAL TONE	50-140
+ 140 V	LOCAL TONE	94-120

3.8.3 Power Supply PP-764( )/G.- The transmitter high voltage power supply shall have the following characteristics:

3.8.3.1 Regulation and Ripple Checks vs. Line Voltage - The power supply shall be capable, after initial setup, of delivering + 750 as read on test meter V2 when the line voltage is adjusted to 115V. With Line voltage adjusted to 126.5 Volts and 103.5 Volts the following ripple limits shall be met.

Switch A	Supply	Limits
1	P.A.	12 volts max.
2	Beam	25 Millivolts max.
3	Refl.	120 Millivolts max

3.9 Receiver Rack.- The Receiver Rack consists of one Radio Receiver R-418( )/G one Order-Wire Receiver R-543( )/TRC-29, one Power Supply PP-690 ( )/G, and one Power Supply PP-689( )/G, in one Electrical Standardized Components Case CY-1280( )/U.

3.9.1 Radio Receiver R-418( )/G - The Radio Receiver shall have the following characteristics:

3.9.1.1 Ripple Voltages of Power Circuit.- Connect AC VTVM to each of the following points. (Check at 103.5 and 126.5V line only).

Voltage	AC VTVM Connection Between	Ripple Limits
+ 250	TB3A, Term 2 & Gnd	7 MV Max
+ 150	TB3A, Term 3 & Gnd.	5 MV Max.
-300	TB3A, Term 5 & Gnd	155 MV Max
-625	TB3A, Term 7 & Gnd.	165 MV Max.
-475	TB3A, Term 6 & Gnd.	165 MV Max.

3.9.1.2 Selectivity.- Set up frequency meter calibration using a check point nearest 120 MCS.

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After this is accomplished, tune frequency meter below and above 120 MCS, until a beat is observed on an oscilloscope, then to determine the 3 DB bandwidth, subtract the beat frequency, and the 3 DB bandwidth shall be 15.0 MCS minimum. Compute center frequency as f1 + f2. Allowable is 120 ± 1 MC. The 6 DB bandwidths are determined as stated above, and shall be a maximum of 20.5 MCS.

3.9.1.3 Dynamic Limiting.- The noise level voltage shall be a maximum of 600 millivolts. After determining the noise level, divide the value obtained by 10 (20db), after adjusting an R.F. generator to this calculated voltage, the maximum shall be 550 microvolts.

3.9.1.4 Residual Noise Measurement.- The residual noise shall be a maximum of 1.80 mv.

3.9.1.5 Metering.- After initial warmup and stabilization with all test equipment removed; rotate switches as indicated below meeting the following limits:

Right Hand Meter Position			Allowable
AFC 1st MULT GRID			30-145
3rd MULT GRID			30-145
AFC IF LEVEL			10-100
IF MIXER			50-90
+ 150 V			95-115
+ 250 V			95-115

Left Hand Meter Position			Allowable
2ND MULT GRID			30-140
AFC MIXER	for	Channel 1	40-130
	Reference	12	30-130
	only	20	30-130
		27	20-130
AFC CORRECTION +			0 ± 5
AFC CORRECTION -			0 ± 5
IF LEVEL			10-40
REFL. V.			60-145
BEAM CUR.			85-110
BEAM V.			90-105

Rotate left hand switch to WAVEMETER position, then time for a peak and reading shall be between 30 and 140.

3.9.1.6 Local Oscillator Tuning Range - With an external wavemeter connected in place of receiver wavemeter and the three local oscillator probes adjust to mid positions, tune wavemeter for peak to meet the following limits: at line voltage of 126.5V frequency shall be below 1698 MCS, at line voltage of 115V wavemeter should be adjustable to 2523 MCS.

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3.9.1.7 Receiver Channel Frequencies.-

Channel No.	Carrier Frequency	Local Oscillator Frequency	Crystal Frequency
1	1715	1835	45.013
2	1745	1865	45.763
3	1775	1895	46.513
4	1805	1925	47.263
5	1835	1955	48.013
6	1860	1980	48.638
7	1885	2005	49.263
8	1910	2030	49.888
9	1935	2055	50.513
10	1960	2080	51.138
11	1980	2100	51.638
12	2000	2120	52.138
13	2020	2140	52.638
14	2040	2160	44.281
15	2065	2185	44.802
16	2090	2210	45.323
17	2115	2235	45.844
18	2140	2260	46.365
19	2165	2285	46.885
20	2190	2310	47.406
21	2215	2335	47.927
22	2245	2365	48.552

3.9.2 Receiver Order Wire R-543/G.- The Receiver Order Wire shall have the following characteristics:

3.9.2.1 Ripple and Regulation.- With the variac adjusted to 103.5 V AC and 126.5V AC respectively, the ripple voltage shall be 70 millivolts or less.

3.9.2.2 O.W. Carrier Trap 6 MC Rejection.- The O.W. Carrier Trap 6 MC Rejection at the ouput test point shall be 5 millivolts or less as seen on a VTVM.

3.9.2.3 Maximum Audio Output.- The maximum Audio Output shall be 3.0 volts or more when tested at the two LISTEN test points, and 2.5 volts at the two THRU AUDIO test points.

3.9.2 4 Audio Level Meter.- The audio level meter after proper setup and tuning shall have a reading between 70 and 130.

3.9.2.5 Frequency Response.- When tested as specified in 4.1.1.5, the frequency response of the Order Wire Receiver Amplifier shall meet the following limits:

	Listen Test Point Voltages
300 cycles	Between 1.3(x) and 0.6 (x)
1000 cycles	Value X (Reference)
3000 cycles	Between 1.3(x) and 0.7 (x)

(NOTE: Value X is actual reading obtained during test. This reading should fall between 1.12 and 1.78.)

3.9.2.6 Signalling.- After proper setup, when appropriate switches and buttons are thrown or depressed the appropriate light shall light, or proper bell and/or buzzer shall sound.

3.9.2.7 Transfer Control Circuit Check.- The appropriate lights shall light, and bell or buzzer shall sound

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3.9.2.8 Carrier Keying Voltage and Buzzer Check.- Interruption of the 6 MC carrier and with an oscilloscope connected at the two KEYING PULSE TEST points should show an amplitude of 20 volts min. And with the carrier not interrupted the amplitude shall be 15 volts min.

3.9.2.9 Diversity Circuit Check - At the DIVERSITY -105 V test point, the DC voltage shall be between -103 V and -114 V. At the DC test point it shall be between -20 and -29 V. As seen on an AC VTVM connected to the MAIN AUDIO test points, the readings shall be between 1.5 and 2.0 volts.

3.9.2.10 Handset Audio Output Level.- With DIV. LIGHT SW and REMOTE HANDSET SW to ON position, there shall be a 1000 cps tone, and by pressing the DIVERSITY CONTROL button, no tone shall be available. During normal conversation, the AC VTVM peaks shall be between 0.3 volts and 0.8 volts approximately. Then connect VTVM to TALK test point and the push to talk switch in ON position with the DIV LIGHT and REMOTE HANDSET switch off, then the VTVM reading shall be less than 20 millivolts. Above requirements are to be met with modulation and without modulation on the Audio Handset Line.

3.9.2.11 Meter Readings.-

Service Selector Position	Meter Selector Postion	Allowance
Normal	O.W. CARR Level	35-140
	DISC +	0 ± 20
	DICS -	0 ± 20
	+ 140 V	95-115
	DIV CONT LEVEL	75-130
	DIV OPER	95-120

(with DIV CONT LEVEL Switch depressed, allowable is 40-100, and with DIV OPER switch depressed allowable is about Ø.)

3.10 Radio Set, AN/TRC-29 System Test.- With all the equipment hooked up as shown in Figure 1, and with Transmitter and Receiver properly tuned, the following requirements shall be met.

3.10.1 System Signal to Noise at 115 Volt Line (80 db Path).- The Receiver output as measured with a VTVM at
the Video Output jack on the O.W. Receiver shall be a maximum of .007 volts (7 millivolts) (Equivalent to S/N of - 37 db max.)

3.10.2 Test Facility Cable Check.- The 20 db test facility cable shall be intact upon the completion of specified test.

3.10.3 System Video Response.- The Receiver output allowable ratio is .70 to 1.12 (-3 db to + 1 db.)

3.10.4 Order Wire Level.- The O.W. CARRIER LEVEL shall be 40-80 as seen on the right hand meter on the transmitter.

3.10.5 Handset Signalling Unit Operation.- When tested as specified in 4.12.5 the O.W. Receiver buzzer shall sound and the push to talk switch on the handset must operate properly with conversation possible in one direction.

3.11 Vibration. -**MISSING WORDS** Except for internal resonance of specified parts and subassemblies, the equipment shall have no mechanical resonance below 55 cycles per second.

3.12 Bounce and shock - The equipment shall meet the requirements of table I

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Table I. Bounce and Shock

Requirement	Inspection Paragraph	Performance after Inspection
Bounce; heavy equipment	4.17	Full specification performance. No physical damage.
Shock; drop	4.14	Equipment shall be operable. Minor mechanical damage is permissible.

3.13 Preconditioning.- Items shall be capable of meeting the inspection of Section 4, without subsequent processing, after subjection, to the bounce preconditioning of 4.4.

3.14 Interchangeability.- Corresponding units and replaceable assemblies, subassemblies, and parts on contract shall be physically and functionally interchangeable as complete items without modification thereof or of other articles with which the items are used. When dimensions, ratings, characteristics, etc., are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular assembly, subassembly, or part is to be considered replaceable, the contracting officer shall be consulted.

3.15 Preproducation samples.- The contractor shall furnish preproduction samples for approval as required by the invitation for bids and contract.

3.16 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order.

3.17 Workmanship.- The equipment shall be manufactured and assembled in a thoroughly workmanlike manner in accordance with the applicable portions of the following paragraphs:

(a) In Specification MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire and cable
- Flux and cleaning agents for soldering
- Process for soldering

Cleaning of equipment
- Riveting
- General requirements for securing parts.
- Cleaning of welding.
- Process for welding.
- Mechanical operation of controls.
- Gears
- Self-tapping screws

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection.- Inspection shall be classified as follows:

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(a) Preproduction inspection (does not include preparation for delivery.) (See 4.2)
(b) Procurement inspection. (Procurement inspection shall be the inspection performed by the contractor and by the Government, as specified by 4.3)

(1) Procurement inspection of items before preparation for delivery. (See 4.3)
(2) Procurement inspection of preparation for delivery. (See Section 5).

4.2 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table II and the group A, group B, and group C inspection specified in tables III, IV, and V respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, and (3) shock, drop; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table II - Preproduction Inspection

INSPECTION	RQT PAR	INSP PAR
Vibration	3.11	4.15
Temperature	3.7	4.6
Shock; Drop	3.12	4.14
Bounce	3.12	4.17
Relative Humidity	3.7	4.13

4.3 Procurement inspection of items before preparation for delivery.- The minimum inspectionperformed by the contractor shall be as specified by (a) through (d) below--to demonstrate compliance with specified requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government - at its discretion--will verify, by product inspection, the contractor's compliance with specified requirements. Such verification shall not exceed the amount of inspection required herein of the contractor.

(a) Procurement inspection shall consist of group A, group B, and group C inspection as specified in 4.3.1 through 4.3.3.2.
(b) When Standard MIL-STD-105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.
(c) Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection, In addition, group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied when it is considered more practical to select seperate samples for group B or group C inspection, or both.
(d) Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly. (See 3.13.)

4.3.1 Group A inspection.- This inspection (including sampling) shall conform to table III and Standard MIL-STD-105. Unless otherwise specified, normal inspection shall be used at the start of the contract. Group A inspection shall be performed in any order which is satisfactory to the Government.

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TABLE III - GROUP A INSPECTION

INSPECTION	RQT PARA.	INSP. PARA.	AQL - dphu **	
			MAJOR	MINOR
Visual and Mechanical:	3.17	4.16		
Radio Transmitter T-303( )/G	3.17	4.16	6.5	25.0
Order Wire Transmitter T-389( )/TRC-29	3.17	4.16	4.0	15.0
Power Supply PP-764( )/G	3.17	4.16	4.0	15.0
Radio Receiver R-418( )/G	3.17	4.16	6.5	25.0
Order Wire Receiver R-543( )/TRC-29	3.17	4.16	6.5	25.0
Power Supply PP-690( )/G	3.17	4.16	4.0	15.0
Power Supply PP-689( )/G	3.17	4.16	4.0	15.0
Test Facility Cable	3.10.2	4.16	1.0	4.0
Transmitter T-303( )/G:				
Tuning	3.8.1.1	4.7.1)		
Modulator Gain	3.8.1.2	4.7.2)		
Modulator Response	3.8.1.3	4.7.3)	6.5	
Monitor Receiver Output Level	3.8.1.4	4.7.4)	for *	
Sawtooth Linearity	3.8.1.5	4.7.5)	the	
Output Power	3.8.1.6	4.7.6)	group	
AFC Hunting	3.8.1.7	4.7.7)		
Monitor Receiver Video Response	3.8.1.8	4.7.8)		
Order Wire Level	3.8.1.9	4.7.9)		
Metering	3.8.1.10	4.7.10)		
Transmitter Order Wire T-389( )/G:				
Frequency drift vs line	3.8.2.1	4.8.1)		
Ripple Voltage vs Line Voltage	3.8.2.2	4.8.2)		
Deviation O.W. Transmitter (R.F.)	3.8.2.3	4.8.3)	2.5	
Output Voltage	3.8.2.4	4.8.4)	for *	
Frequency Response of Audio Amplifier	3.8.2.5	4.8.5)	the	
Alarm Circuits	3.8.2.6	4.8.6)	group	
Metering Circuits	3.8.2.7	4.8.7)		
Radio Receiver R-418( )/G:				
Ripple Voltage of Power Circuit	3.9.1.1	4.10.1)	6.5	
Selectivity	3.9.1.2	4.10.2)	for *	
Dynamic Limiting	3.9.1.3	4.10.3)	the	
Metering	3.9.1.4	4.10.5)	group	
Local Oscillator Tuning Range	3.9.1.5	4.10.6)		
Receiver Order Wire R-543( )/G:				
Ripple and Regulation	3.9.2.1	4.11.1)		
O.W. Carrier Trap GMC Rejection	3.9.2.2	4.11.2)		
Maximum Audio Output	3.9.2.3	4.11.3)		
Audio Level Meter	3.9.2.4	4.11.4)	4.0	
Frequency Response	3.9.2.5	4.11.5)	for *	
Signalling	3.9.2.6	4.11.6)	the	
Transfer Control Circuit Check	3.9.2.7	4.11.7)	group	
Carrier Keying Voltage & Buzzer Check	3.9.2.8	4.11.8)		
Diversity Circuit Check	3.9.2.9	4.11.9)		
Meter Readings	3.9.2.11	4.11.11)		

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TABLE III (Cont'd)

INSPECTION	RQT PARA.	INSP. PAR	AQL - DPHU MAJOR	AQL - DPHU MINOR
Radio Set AN/TRC-29 System Test:				
System Signal to Noise at 115V Line	3.10.1	4.12.1)		
Test Facility Cable Check	3.10.2	4.12.2)	2.5 for the	
System Video Response	3.10.3	4.12.3)	Group *	
Order Wire Level	3.10.4	4.12.4)		
Handset Signaling Unit Operation	3.10.5	4.12.5)		

*All electrical defects are considered MAJOR
** DPHU = Defects per hundred units

4.3.2 Group B inspection.- This inspection (including sampling) shall to table IV and the Appendix to Standard MIL-STD-105. Unless otherwise specified herein normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with 4.3.5 and the requirements of Standard MIL-STD-105 for disposition of rejected product.

4.3.2.1 Group B sampling plans.- The group B sampling shall be in accordance with the plan indicated below:

Group B Plan	AQL	Inspection level for normal inspection	Inspection level for reduced inspection
B-2	6.5%	L-7	L-5

4.3.2.2 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

4.3.2.3 Procedure in case of failure.- When an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

Table IV - Group B Inspection

Inspection	RQT PAR	INSP PAR	Group B Plans (See 4.3.2.1)
Power Supply PP-764( )/G:			
Regulation & Ripple Check vs Line Voltage	3.8.3.1	4.9.1	B-2
Radio Receiver R-418( )/G:			
Residual Noise Measurement	3.9.1.4	4.10.4	B-2
Receiver Order Wire R-543( )G:			
Handset Audio Output Level	3.9.2.10	4.11.10	B-2
Interchangeability	3.14	4.18	

4.3.3 Group C inspection.- This inspection shall be as listed in table V.

4.3.3.1 Sampling for inspection of equipment.- As specified by Specification MIL-I-11748.

4.3.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall

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immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

TABLE V - GROUP C INSPECTION

INSPECTION	RQT PAR	INSP PAR
Radio Interference Suppression	3.6	4.5.MISSING

4.3.4 Reinspection of conforming group B and group C Sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.3.5 Disposition of non conforming product.- The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault:

Sample units found defective during group A, B, or C inspection.
Inspection lots which failed during group A or B inspection.
Product which has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.4 Bounce preconditioning.- The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection of procurement inspection.

4.5 Radio Interference suppression.- Inspection for suppression shall be performed in accordance with Specification MIL-I-11748.

4.6 Temperature test for equipment.- The equipment shall be subjected to the temperature cycle shown on MIL-STD-169.

4.7 Transmitter T-303( )/G.- Following are tests for Transmitter T-303( )/G:

4.7 1 Tuning - The transmitter tuning shall be capable of meeting the full requirements of 3.8.1.1.

4.7.2 Modulator Gain.- Throw BIAS switch on H.V.P.S. to OFF. Remove cage from output side of modulation. Connect a probe from a VTVM between ground and the junction of C-25, high voltage coupling capacitor (0.1 uf, 3000V) and the 10 ohm resistor R-52. Turn VOLTAGE RANGE control on VTVM to 150.. Throw BIAS switch to ON and adjust line voltage to 103:5 volts. Adjust the frequency of a signal generator to 500 cps and set its level for a 1.2 volt RMS reading on VTVM connected to generator output terminals. Due to the dividing action of the 560 ohm resistor and the 50 ohm terminating resistor, in the modulator, the level at the modulator input is 0.1 volts RMS. Switch the modulator input selector switch to VID. The output shall meet the requirements of 3.8.1.2.

4.7.3 Modulator Response.- Adjust generator frequency to 100 KC. Adjust generator level for a reading of 1.2 volts on VTVM. Adjust generator frequency to 2.5 MCS. Set generator level at 1.2 V. Note reading on VTVM. The requirements of 3.8.1.3 shall be met. Adjust generator frequency to 7.0 MCS. Set generator level to 1.2 volts as read on G. R. VTVM, M-1. The requirements of 3.8.1.3 shall be met.

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4.7.4 Monitor Receiver Output Level.- Connect a VTVM across a 50 ohm resistor at the MON. REC. OUT. jack. Adjust the level of a H.P. 650A, or equal, to obtain a 100% modulation on peak VTVM. The readings obtained shall meet the requirements of 3.8.1.4.

4.7.5 Sawtooth Linearity.- Switch MOD. INPUT SEL SW. to ST position and VTVM SEL. SW. to MON. OUT position. Adjust ST LEVEL control, near top of VTVM chassis, for a reading of 100% modulation on PEAK VTVM meter in NEG PEAK position. Sawtooth voltage is now adjusted to give 3.6 MC deviation. Observe sawtooth pattern on Tektronix 514D scope connected to MON. OUTPUT jack.

4.7.6 Output Power.- Switch MOD INPUT SEL SW to GND. Check that transmitter is connected thru attenuator box and bolometer to PRD bridge. Zero bridge before taking a reading. Check to see that P.A. CATH CURR is 100. Readjust variac on H.V.P.S. if necessary. Adjust line voltage to 115 volts. Obtain a power reading on power bridge and calculate the true power using the calibration of the RG-21/U cable.

4.7.7 AFC Hunting.- Throw MANUAL FREQUENCY switch to INCREASE position and hold in this position until the left hand meter reads approximately 30, then release the switch. The meter return shall meet the requirements of 3.8.1.7.

4.7.8 Monitor Receiver Video Response.- Feed a signal generator set at 100 KC into the modulator with the MOD INPUT SEL. SW. to VID and VTVM SEL. SW. to MON. OUT. Adjust generator output level for 20 volts at the modulator output. Note meter reading of PEAK VTVM in NEG. PEAK position. This is the 100 KC reference. Set signal generator to 300 KC with 20 volts output from the modulator. Note reading. Set signal generator for 4 MC with 20 volts output from modulator. Note reading. The requirements of 3.8.1.8 shall be met

4.7.9 Order Wire Alarm.- Connect a 150 ohm resistor in series with a signal generator to ORDER WIRE CARRIER INPUT jack. Turn MOD INPUT SEL SW to GND. Set signal generator frequency to 1.5 MCS and adjust level to obtain 100% modulation on VTVM in MON OUT, NEG PEAK position. Note VTVM reading. Set signal generator frequency to a calibrated 6 MCS and adjust generator output to obtain .15 (15%) of the above reading. The lamp on meter panel of transmitter marked TRANSMITTER OPERATING should be lit. Slowly reduce generator level until lamp just goes out. Note the voltage reading on VTVM and meet the requirements of 3.8.1.9.

4.7.10 Metering.- Rotate metering switches as indicated in 3.8.1.10 noting that all reading meet the requirements of 3.8.1.10.

4.8 Order Wire Transmitter T-389( )/TRC-29 - Following are tests for Order Wire Transmitter:

4.8.1 Frequency Drift vs Line.- With the variac adjusted for 115 V and the AUDIO SELECTOR switch set to the LOCAL position, measure the carrier frequency using a frequency counter. Repeat the above measurement at 103.5 V and at 126.5 V. The requirements of 3.8.2.1 shall be met.

4.8.2 Ripple Voltage vs Line Voltage.- With PLATE circuit breaker off, disconnect VTVM from S1 and connect an AC VTVM in its place. With circuit breaker on, adjust VARIAC to 103.5V, 126.5V, and 115 V respectively. Note readings on VTVM. The requirements of 3.8.2.2. shall be met.

4.8.3 Deviation O.W. Transmitter RF.- Connect 1 megacycle beating signal from crystal calibrator to Panalyzer, and the Panalyzer to R-F output. Set AUDIO SELECTOR switch to THRU position. Adjust audio oscillator for an output of 3 volts RMS at 30.8 KC. Connect AC VTVM between V2 pin 7 and ground. Adjust R-1 AUDIO LEVEL THRU CHANNEL to obtain a reading of 0.3 V RMS on AC VTVM. Adjust L-2 MOD SENS ADJ for first carrier disappearance on Panalyzer. The requirements of 3.8.2.3 shall be met.

4.8.4 Output Voltage.- Turn R-1, AUDIO LEVEL THRU CHANNEL, and R-2 AUDIO LEVEL LOCAL INPUT fully counter-clockwise, then connect AC VTVM to output and connect cable from ORDER WIRE CARRIER OUTPUT to CARRIER OUT on test. The readings of AC VTVM shall meet the requirements of 3.8.2.4.

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4.8.5 Frequency Response of Audio Amplifier.- With a 1000 cps signal of .776 V applied to the THRU input of the order wire transmitter, resistor R1 adjusted to deliver 0.3 V RMS to a VTVM connected between pin 7 of T1 and ground, and Resistor R 24 adjusted for a reading of 100 on Meter M1. After the above check of the THRU input switch to the LOCAL input and apply a 1000 cps signal of .776 V, then adjust R2 to deliver 0.3V RMS to the VTVM. After the above is accomplished the Frequency Response shall meet the requirements of 3.8.2.5.

4.8.6 Alarm Circuits.- Depress the 6 MC ALARM and the OFF FREQUENCY ALARM buttons. The requirements of 3.8.2.6 shall be met.

4.8.7 Metering Circuits.- Set AUDIO SELECTOR switch to THRU position, apply 1000 cps with an output of 0.766 volts, then connect AC VTVM to V2 pin 7, adjust R-1 for 0.3 volts as seen on AC VTVM. Switches S-1 and S-2 shall then meet the requirements of 3.8.2.7.

4.9 Power Supply PP-764( )/G.- Following is test for Transmitter High Voltage Power Supply:

4.9.1 Regulation and Ripple Checks vs Line Voltage.- With an input of 115 V, throw the INPUT POWER BREAKER on, also throw switches CB4, CB3, CB5, CB6, and CB7, turn Variable Transformer Variac fully counter-clockwise, then throw CB8 switch, rotate the P.A. VOLTAGE switch to position 1. Adjust the VARIABLE AUTO-TRANSFORMER to meet the requirements of 3.8.3.1. After the above requirement is met, increase line voltage to 126.5 volts, and then decrease line voltage to 103.5 volts. The requirements of 3.8.3.1 shall be met at the high and low voltage conditions.

4.10 Receiver R-418/G.- Following are tests for Radio Receiver:

4.10.1 Ripple Voltages of Power Circuit.- Set AFC switch to OFF position. Turn on POWER, H.V. FIL and L.V. PL breakers, allow a two minute stabilization period. Adjust line voltage to 115 V, then turn up H.V. PL Breaker and allow two minutes more for stabilization, then readjust line voltage. Turn left hand meter switch to BEAM V. position, release each power supply and receiver interlock to check that meter reading drops. Connect DAC VTVM to check points listed in 3.9.1.1 to check ripple. Check at 103.5 V and 126.5 V line voltages to meet requirements of 3.9.1.1.

4.10.2 Selectivity.- Connect Receiver as shown in Figure I. Adjust the R-F generator frequency for crossover with right hand meter switch on IF DISCR. FREQ. LOW. Adjust R.F. generator output level so that the LEVEL METER reads 65. To determine the maximum in the receiver response, slowly tune r-f generator above and below center frequency. Set R-F generator to the frequency that corresponds to the maximum r-f level. Adjust R-F Generator level so that I-F level meter reads exactly 65. Remove one 3 db pad and reconnect cable. Recheck the I-F level. Slowly tune frequency meter below 120 mcs until a beat is seen on oscilloscope keeping scope vertical gain near maximum. Note frequency meter reading as f1. Check I-F level again for 65 then tune frequency meter above 120 MCS until a beat is seen on scope. Record this reading as f2. Then determine 3 DB bandwidths and center frequency per 3.9.1.2. Then repeat above procedures and determine the 6 DB bandwidths to comply with 3.9.1.2.

4.10.3 Dynamic Limiting.- Connect an A.C. VTVM to VIDEO OUPUT MONITOR jack across 51 ohm resistor. The R-F- generator should remain connected to receiver input (see figure 2). Turn switch on signal generator from CW to OFF and note VTVM noise level reading. This voltage shall meet the requirement of 3.9.1.3. Turn signal generator switch to CW and set frequency to crossover. Then divide by 10 (20 db) the above voltage on VTVM. Note generator attenuator setting, thus meeting the requirements of 3.9.1.3.

4.10.4 Residual Noise Measurement.- Set generator level to the point where video level no longer decreases with increasing carrier level (as seen on AC VTVM). It is permissable to retune signal generator to one of peaks of discriminator to obtain minimum residual noise reading.

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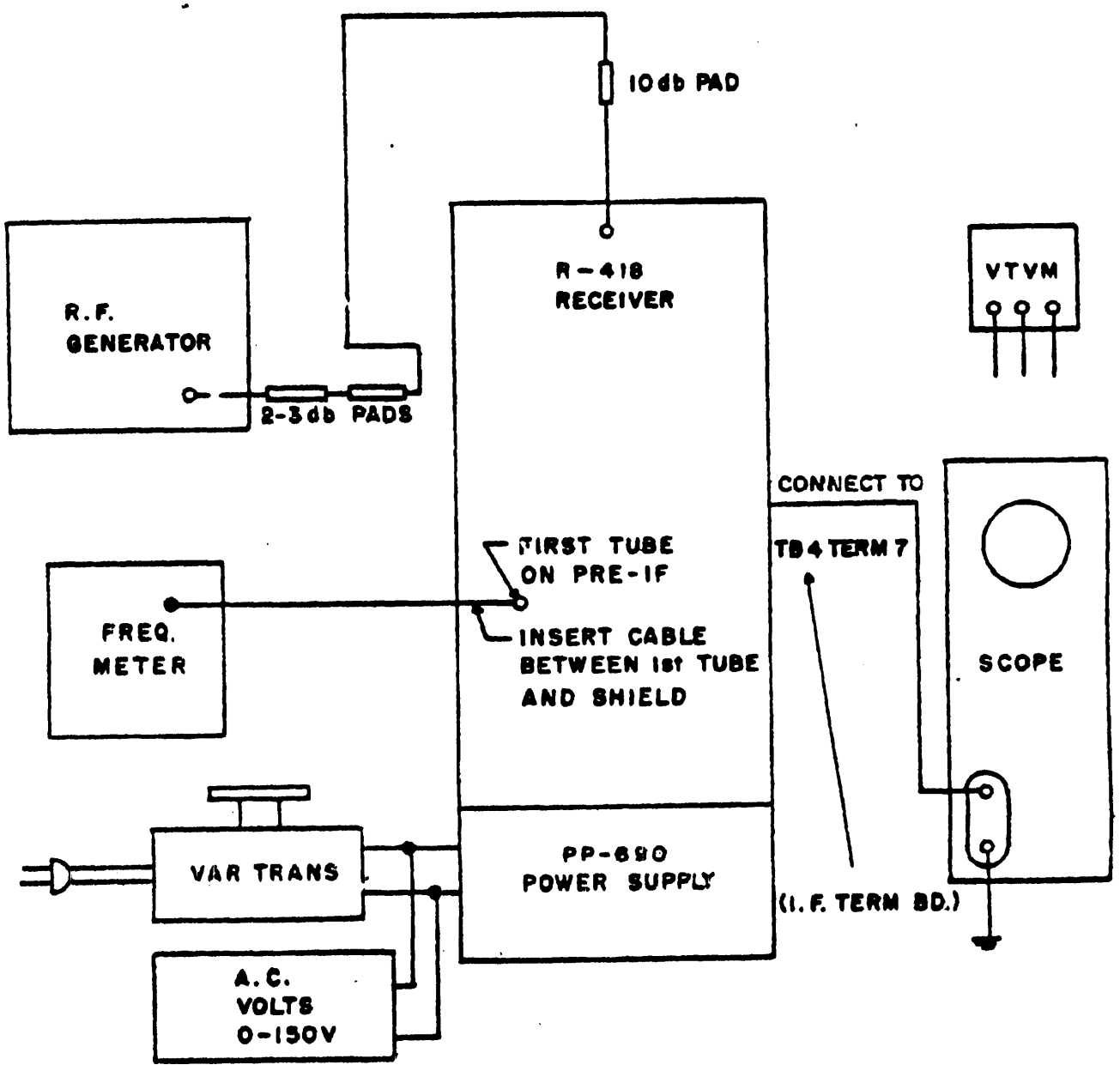


FIGURE 1

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4.10.5 Metering.- Disconnect signal generator from receiver input, and with all other test equipment removed from receiver, rotate meter switches on front panel to positions as indicated in 3.9.1.5.

4.10.6 Local Oscillator Tuning Range.- Connect an external calibrated wavemeter in place of receiver wavemeter and turn meter switches to WAVEMETER and I.F. MIXER position. Set AFC switch to OFF, while turning TUNING KNOB to maximum CCW positions. Adjust three local oscillator probes to their mid positions. Raise line voltage to 126.5 V. Turn REFL V. CCW until a maximum I.F. MIXER reading is obtained. (REFL V. should be near low end of range for best operation). Tune wavemeter for a peak. Wavemeter should indicate a frequency below the requirement of 3.9.1.6. Reduce line voltage to 115 V, the adjust Wavemeter to the requirements of 3.9.1.6.

4.11 Receiver Order Wire R-543/G.- Following are tests for the Receiver Order Wire:

4.11.1 Ripple and Regulation.- Disconnect DC voltmeter and connect an AC VTVM in its place, then adjust variac to 103.5 V, 115 V, and 126.5 V, note readings of AC VTVM.

4.11.2 O.W. Carrier Trap 6 MC Rejection.- Adjust signal generator frequency (with modulation off) to 6.000 MCS using crystal calibrator, then adjust generator output to 50 millivolts. Connect Ballantine AC VTVM between V-1, pin 1 and ground. Note VTVM reading.

4.11.3 Maximum Audio Output.- F.M. Modulate signal generator ± 50 KCS at 1000 cps using external audio oscillator, turn R 26 AUDIO LEVEL fully CW, connect AC VTVM to LOCAL AUDIO test point, then note reading. Connect the AC VTVM to the two LISTEN test points (REMOTE HANDSET switch should be OFF). Note readings. Then connect the AC VTVM to the two THRU AUDIO test points. Note readings. All readings shall meet the requirements of 3.9.2.3.

4.11.4 Audio Level Meter.- Adjust R26 for a reading of 0.776 V on the AC VTVM. Set S1, METER SELECTOR switch to AUDIO position, note reading of receiver panel meter.

4.11.5 Frequency Response.- On audio oscillator with an output impedance matching the input of the Receiver Order Wire Amplifier shall be applied with a signal level of 0.776 V at 1000 cycles. The output, at the LISTEN test point shall be between 1.12 V and 1.78 V, and shall be recorded. At 300 cps the input level shall be adjusted to 84 V, and at 3000 cps, the output level shall be adjusted to .73 V. The output, at the LISTEN test point, at these frequencies, shall meet the requirements of 3.9.2.5.

4.11.6 Signalling.- Throw DIV. REC. LIGHT switch and REMOTE HANDSET switch to on. Set 32 BUZZER CONTROL switch to MONITOR position; the requirements of 3.9.2.10. shall be met. Reset S2 to NORMAL position. Depress S5 RING switch; the requirements of 3.9.2.10 shall be met. Press REMOTE RING button, the requirements of 3.9.2.6 shall be met Press REPEATER RING button, then press DIV. RCVR RING button, both above pressings shall meet the requirements of 3.9.2.6.

4.11.7 Transfer Control Circuit Check.- Switch S3 SERVICE SELECTOR switch to TRANSFER CONTROL, as specified in 3.9.2.7 the appropriate lights shall light. Press ALARM CODER button, press S5 RING button, throw RING DISABLE switch to ON and press S5 RING button, at each of the above actions, the requirements of 3.9.2.7 shall be met.

4.11.8 Carrier Keying Voltage and Buzzer Check.- Connect the oscilloscope to the two KEYING PULSE test points and the KEYING PULSE OUT test point, with the oscilloscope at each of these points, interrupt the 6 MCS carrier by disconnecting the cable from J1, VIDEO INPUT FROM REC jack. A pulse should appear on the scope at the instant the 6 MC signal is interrupted, note and record the amplitude of the pulse. Then reconnect the cable to jack J1, connect the scope to the KEYING PULSE OUT test point, press S5 RING button, (with RING DISABLE SW ON). A pulse should appear on the scope when S5 RING button is pressed Note and record the amplitude of the pulse for compliance with 3.9.2.8.

4.11.9 Diversity Circuit Check.- Set SERVICE SELECTOR switch to DIVERSITY position. Press DIVERSITY RECEIVER ALARM switch, Buzzer and bell should sound, DIV RECEIVER BELL lamp and DIV RECEIVER BUZZER lamp should light, then release DIV RECEIVER ALARM button; connect DC Voltmeter to the DIVERSITY -105V test point. Record this reading, then press DIVERSITY CONTROL switch, voltmeter reading should drop off to zero. Connect DC Voltmeter to the DC test point, note and record voltmeter reading. Then disconnect voltmeter and connect signal generator lead to INPUT jack, MODULATE the signal generator ± 50 KC at 1000 cps. Connect the AC VTVM to the MAIN

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AUDIO test points, record the VTVM readings, then press DIVERSITY CONTROL button, again note and record the reading. All of the above readings shall meet the requirements of 3.9.2.9.

4.11.10 Handset Audio Output Level - With DIV. LIGHT SW. and REMOTE HANDSET SW to the ON position, listen to phone on front of receiver, there shall be a 1000 cps tone audible. With the DIVERSITY CONTROL button depressed, there is no tone. With no external modulation, connect an AC VTVM to the three test points in turn, then speak, in a normal conversation, into the handset. Record the peak readings. (Note: Peaks vary with individual voices.) Then connect the VTVM to any TALK test point and record the VTVM reading with DIV LIGHT switch and REMOTE HANDSET switch OFF, and the push-to-talk handset switch ON. Upon completion of the above repeat all of the above tests with an external modulation, thus meeting the requirements of 3.9.2.10.

4.11.11 Meter Reading.- With all test equipment removed from the order wire Receiver, rotate meter switches to positions indicated in 3.9.2.11.

4.12 Systems (Radio Set AN/TRC-29( )).- (See Figure 3)

4.12.1 System Signal to Noise at 115 Volt Line (80 db Path).- The system signal to noise is the ratio of the receiver video output when receiving an FM signal modulated + or - 4.5 mcs to the noise when the input to the transmitter is grounded. Set input voltage to transmitter and receiver at 115 volts. Set controls as follows: O.W. Trans. Plate off, O.W. Rec Buzzer Control to Monitor, Transmitter Mod. Input Sel Switch to Vid, Transmitter VTVM Sel Switch to Mon. Out, Transmitter Pos. Peak - Neg. Peak switch to Neg. Peak. Connect a signal generator through a 560 ohm resistor to VIDEO INPUT jack at rear of transmitter. Turn modulator INPUT VIDEO LEVEL pat. fully CW. Set generator frequency to 1.87 mcs and output to obtain a reading of 62.5 on black scale of VTVM. Connect a Ballantine 314, or equal, across 50 ohm on VIDEO OUTPUT jack of O.W. REC. Adjust VIDEO LEVEL pat. on Receiver I.F. for a reading of 0.5 volts on Ballantine. Turn transmitter MOD INPUT SEL SW to GND Using a Ballantine 314, or equal, the measurements shall meet the requirements of 3.10.1.

4.12.2 Test Facility Cable Check.- To prevent P.A. arc over while disconnecting cables, throw BIAS switch on transmitter to OFF, disconnect lossy cable from transmitter and add 20 db test facility cable. Throw BIAS switch to on. Connect signal generator to VIDEO INPUT jack on transmitter and VTVM to VIDEO OUTPUT jack. Set generator frequency to 10 KC and output to a minimum. Increase generator output. Receiver video output should also increase, thus meeting the requirements of 3.10.2.

4.12.3 System Video Response.- Using the same test set-up as for Test Facility Cable Check. Check system response at 30 cps, and 4.5 mcs. Reference reading will be taken at 100 KC, input to transmitter being held constant. Throw BIAS switch of high voltage supply to OFF. Remove VTVM from modulator output and connect it to VIDEO INPUT jack. Replace cage on modulator and throw BIAS switch to ON. Check to see that MOD INPUT SEL SW is in VID position and that VIDEO LEVEL pot is fully CW. Set generator frequency dial to 10, multiplier to X10 KC and level to obtain .05 volt on VTVM. Record this reading at receiver output. This is 100 KC reference for low end. Change generator frequency to 30 cps and adjust level for .05 volts on input VTVM. Note receiver output on VTVM Calculate ratio of (30 cps. Reading)
(100 KC Reference) for low end.
Set generator frequency dial to 1, multiplier to X100 KC and level to obtain .05 volts on input VTVM. Note and record the VTVM reading at the receiver output, this is 100 KC reference for high side. Change generator frequency to 4.5 mcs and level to obtain .05 volts on input VTVM. Note VTVM reading at receiver output. Calculate ratio of (4.5 mc Reading)
(100 KC Reference) for high end.
The above two calculations shall meet the requirements of 3.10.3.

4.12.4 Order Wire Level.- The proper order wire carrier level is such that the R.F. carrier is deviated approximately 20% in the absence of other modulation. Set the controls as follows: O.W. Transmitter Plate to ON, O.W. Carrier Level Pot fully CCW, Transmitter Mod. Input Sel Sw. to VID, Transmitter VTVM Sel Sw to Mon OUT, Transmitter Pos. Peak-Neg Peak Sw to Neg Peak. Adjust signal generator frequency to 1.5 MCS and level to obtain 100% modulation on Peak VTVM. Set VTVM Sel Sw to MOD OUT and record reading on PEAK VTVM. Turn MOD INPUT SEL SW to GND. Adjust O.W. CARRIER LEVEL CONTROL for 20% of above recorded reading. Lock O.W. CARRIER LEVEL pot. Switch right hand meter on transmitter to O.W. CARRIER LEVEL; and the requirements of 3.10.4 shall be met.

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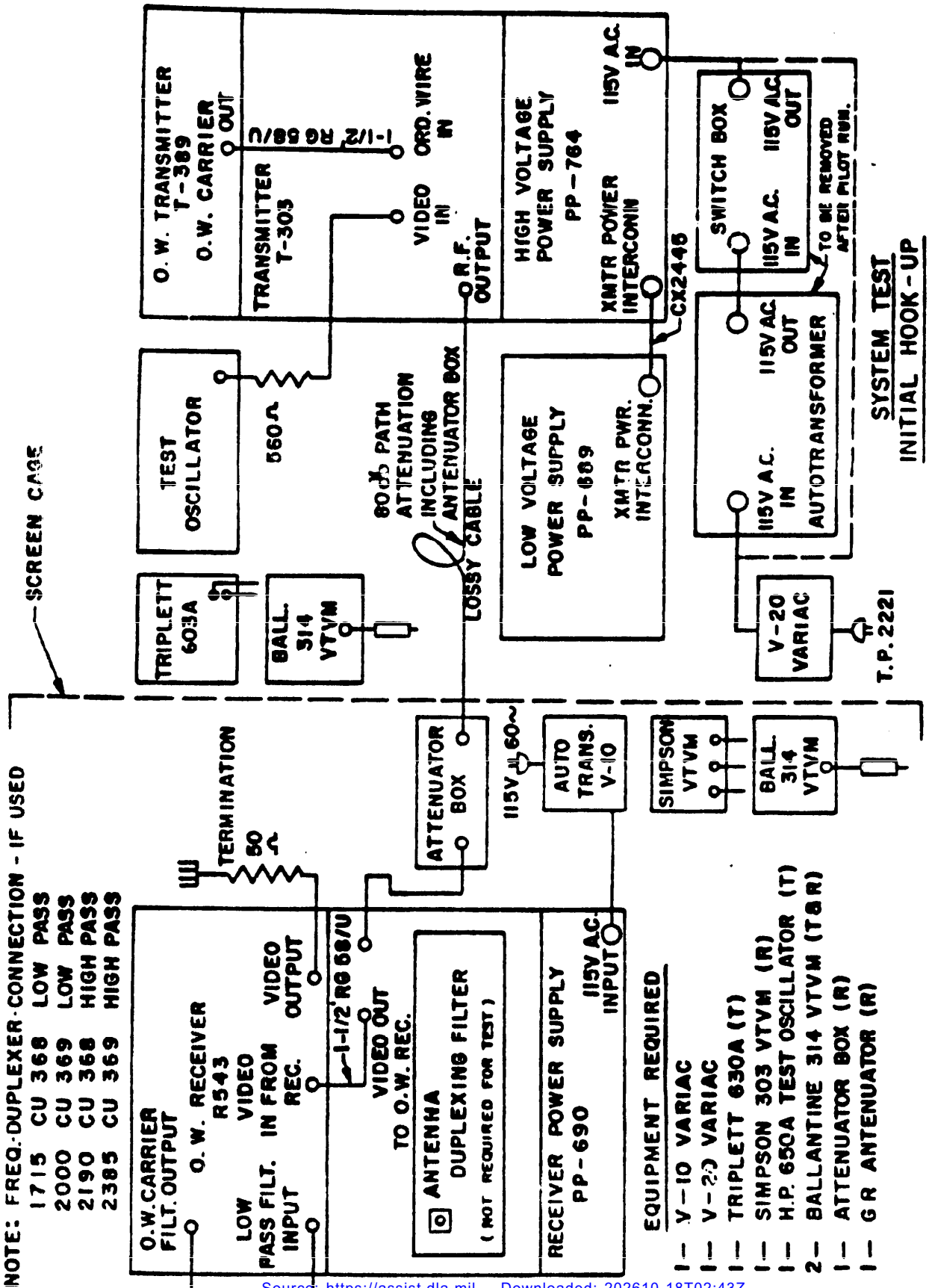


FIGURE 3

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4.12.5 Handset Signalling Unit Operation.- The handset should remain connected to the O.W. transmitter to allow one way conversation through O.W. transmitter and receiver, with the return path via the intercom. Set test switch in O.W. transmitter to operate. Press ring button on O.W. Transmitter, the buzzer should sound. The requirements of 3.10.5 shall be met.

4.13 Moisture-resistance test for equipment.- The equipment shall be tested as follows:

4.13.1 Test Conditions.-

(a) Do not remove equipment from the humidity chamber for measurements.
(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.
(c) Test sealed equipment with the seal intact except during last two cycles. Bring out leads for electrical measurements through a hole in the equipment enclosure and seal against entrance of moisture. Perform last two cycles with the chassis removed from its enclosure.

4.13.2 Test method.-

(a) Dry at 130° ± 5°F for 24 hours.
(b) Condition at 77° ± 5°F and 40 to 50 percent relative humidity for 24 hours.
(c) Take measurements as in 4.13.3, and readjust or reline as necessary to meet full specification requirements.
(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.13.3 shall be made during the periods shown on the Standard and shall comply with 4.13.3.
(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5°F and 40 to 60 percent relative humidity. The adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet the full specified performance for those measurements specified in 4.13.3.

4.13.3 Performance.- Measurements made during test, and required performance during cycling shall be "full specification performance".

4.13.4 Failure.- If any equipment fails to meet the performance specified in 4.13.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.13.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.13.2(e), it does not pass the test.

4.14 Shock test; drop.- The equipment, in its transit or combination case, shall be subjected to the drop test in accordance with table VI. The floor or barrier receiving the impact shall be of 2-inch fir backed by concrete or a rigid steel frame.

Table VI - Drop Test

Weight of Equipment Including Case	Largest Dimension Of Case (inches)	Note	Height of Drop (inches)	Number of Drops
Over 200, through 1000 pounds.	36 to 60	A	36	Drop on each Corner. Total of 8 drops.
1000 pounds.	60 and over	A	24	

NOTE A: With its longest dimension parallel to the floor, the transit or combination case, with the equipment within, shall be supported at the corner of one end by a block 5 inches in height, and at the other corner of the same end by a block 12 inches in height. The opposite end of the case shall then be raised to the specified height and allowed to fall freely.

4.15 Vibration test.- The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.11 as follows:

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(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts, if any, shall be blocked. The vibration table shall provide approximately senusoidal vibration. (See 4.15(d)).

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because the unit weighs more than 250 pounds or inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

4.16 Visual and mechanical inspection. - Equipment shall be examined for the defects listed in Standard MIL-STD-252 and table VII.

TABLE VII - CLASSIFICATION OF VISUAL AND MECHANICAL DEFECTS

CLASSIFICATION	DEFECT
MAJOR	Carrying handle, hinge, or lid fastener attached to wrong side of case. Cover missing or inoperative. Control inoperative
MINOR	Cover reinforcement missing or improper assembly. Control: Difficulty in getting good contact, or in making accurate setting due to play or backlash. Does not operate smoothly and freely
CONTROL	Carrying handle or cover difficult to operate.

4.17 Bounce Test; equipment under 1000 pounds - The equipment, in its transit case, shall be placed on the table of Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 RPM ± 1 percent, for a total of 3 hours. During each ½ hour period of the test, the transit case (with contained equipment) shall rest on a different face.

4.18 Inspection for interchangeability. - The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.14. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mounting

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.19 Procurement inspection of preparation for delivery - The minimum inspection performed by the contractor shall be as specified in 4.19.1 and 4.19.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government - at its discretion - will verify, by product inspection, the contractor's compliance with specified requirements. (See 6.6.) Such verification shall not exceed the amount of inspections by the

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Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.19.1 Preservation and packaging.- Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall be as shown on table VIII.

TABLE VIII

PRESERVATION, AND PACKAGING AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Item not properly blocked or braced within the unit package to prevent movement
2. Quantity in unit package not as specified.	2. Any item of marking information other than 7 thru 9 omitted, incorrect, or illegible.
3. Incorrect packaging method applied.	
4. Cushioning or padding omitted.	
5. Cushioning inadequate for the physical and mechanical protection of the item.	
6. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.	
7. Stock number omitted, incorrect, or illegible.	
8. Nomenclature omitted, incorrect, or illegible.	
9. Marking of quantity of items in package omitted, incorrect or illegible.	
10. Different stock numbered items in the same unit package	

4.19.2 Packing and marking of exterior containers.- Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent. Classification of defects shall be as shown on table IX.

TABLE IX

PACKING, AND MARKING THEREOF

MAJOR	MINOR
1. Use of improper or defective material.	1. Unsealed carton.
2. Quantity in pack not as specified.	2. Defective taping or sealing of carton.
3. Case liner omitted.	3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Case liner improperly sealed, punctured, or damaged.	4. Any item of required marking information other than 10 thru 17 listed under major defects omitted, incorrect, or illegible.
5. Box closure not as specified.	
6. Type, grade, class, and style of the shipping container not as specified.	
7. Strapping omitted.	
8. Strapping inadequate or incorrectly applied.	
9. Shipping documents or packing list omitted.	
10. Stock number omitted, incorrect, or illegible.	
11. Nomenclature omitted, incorrect, or illegible.	
12. Marking of quantity of items in pack omitted, incorrect, or illegible.	
13. Destination marking omitted, incorrect, or illegible.	
14. Service designation (color marking) omitted.	
15. Set or assembly markings omitted.	
16. Specified special marking and labeling such as NAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.	
17. Overseas code marking omitted, incorrect, or illegible.	

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4.19.2.1 Inspection lot.- A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.19.2.2 Procedure in case of failure.- If an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.19.2.3 Disposition of nonconforming product.- Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of Standard MIL-STD-105 for disposition of rejected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.20 Rough handling test (preparation for delivery).- When rough handling test is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling. Metering, Transmitter Rack 4.7.10, 4.8.7, Metering, Receiver Rack 4.10.5, 4.11.11. (See 6.1(e)(2)c)

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Components of Rear Area Radio Relay System shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and as described herein.

5.1.1.1 Technical manual.- Each technical manual shall be packaged Method IC-3 as follows: Inclose each technical manual within a type I, class b bag conforming to Specification MIL-B-117. Heat seal the bag to effect closure.

5.1.1.2 Radio Set AN/TRC-29( ).-

5.1.1.2.1 Tubes (spares).- Spare tubes shall be packaged individually in accordance with the applicable provisions of Specification MIL-P-75.

5.1.1.2.2 Transformer TF-146( )/G.- Each transformer shall be packaged Method III as follows: Cushion each transformer on all surfaces with cells or pads or both fabricated of double faced corrugated fiberboard having a dry bursting strength equal to that of the box in which they will be used. Place the cushioned transformer within a close fitting fiber box, class 1 of Specification PPP-B-636. Box closure shall be in accordance with the requirements of the appendix of the box specification.

5.1.1.2.3 Ancilliary items, (other than Transformer TF-146( )/G) accessories and spare parts.- No packaging required.

5.1.1.2.4 Stowage in cases.- Stow all the ancilliary items, (other than Transformer TF-146( )/G) accessories and spare parts within the compartments of the designated cases. As required, fill all voids with pads fabricated of single face flexible corrugated fiberboard, type III conforming to Specification LLL-F-291. Close all the cases and secure all fastenings.

5.1.3 Level C.- Components of Rear Area Radio Relay System shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2

5.2.1 Level A.- Components of Rear Area Radio Relay System shall be packed for shipment within individual close-fitting style 2, class 2 nailed wood boxes conforming to Specification PPP-B-621. Boxes 1/11 and 2/11 (see 5.2.1.2) shall be modified to include wood skids in accordance with the requirements of the box specification. Box 7/11 (see 5.2 1 2) shall be fitted with a case liner grade B

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type CW-1 conforming to Specification MIL-B-13239, fabricated and sealed in accordance with Specification MIL-L-10547. Place the technical manual (see 5.1.1.1) between the contents of Box 1/11 (see 5.2.1.2) and the lid of the box. Box closure shall be in accordance with the appendix of the box specification.

5.2.1.1 Metal strapping.- Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification only for direct shipment to ports.

5.2.1.2 Boxes containing components of Rear Area Radio Relay System shall be arranged in the following sequence:

Transmitter Rack, An/TRC-29( ) in Case CY-1280/U	Box 1/11
Receiver Rack AN/TRC-29( ) in Case CY-1280/U	Box 2/11
Running spares and accessories, Case Cy-250	Box 3/11
Running spares and accessories, Case CY-250	Box 4/11
Cable and reel assembly	Box 5/11
Radio Frequency Transmission Line Assy-CG-1013( )/U	Box 6/11
Transformer TF-146( )/G	Box 7/11
Antenna feed horn chest Case 1	Box 8/11
Antenna casting chest Case 11	Box 9/11
Antenna right half Case 111 and Antenna left half Case IV	Box 10/11
Antenna, canvas bag Case V	Box 11/11

5.2.2 Level B.- Components of Rear Area Radio Relay System shall be packed for shipment as specified in 5.2.1 except boxes shall be class 1, Specification PPP-B-621. No strapping required.

5.2.3 Level C.- Components of Rear Area Radio Relay System shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification No. 4 and Consolidated Freight Classification Rules 21 for rail shipment, National Motor Freight Classification No. A-3 and No. 14 for truck shipment, and the regulations of other carriers as applicable to the mode of transportation employed at lowest transportation rate.

5.3 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto
(b) Type required.
(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)
(d) The specific paragraphs of section 5 which are applicable to the particular procurement.
(e) Preproduction inspection:

(1) Two smples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample.
(2) Preproduction pack(s) as follows:
   a. Makeup of pack(s).
   b. Number of each kind of pack to be submitted.
   c. Inspection to be performed thereon-including rough handling test, which will not be performed as procurement inspection.
   d. Five copies of process sheet, using format shown on Drawing SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack
   e. A packaging technician from the procuring agency will evaluate the preproduction pack(s).

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(f) Marking and shipping of samples.
(g) Place of final inspection.
(h) Technical literature required.
(i) Quantity of tools and running spare parts required.
(k) Submission of the statement of treatment referenced in 3.4.3, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.2 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/PIQ 1W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.3 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.3.3.2)

6.4 Definition of level B.- Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.5 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.5.1 Examination.- Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.5.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.6 Verification inspection.- The amount of verification inspection (by the Government) will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

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